

Module 15: Population and Systems Ecology — Questions

Quick Check

- 1. A population of rabbits is introduced to a large island with no predators and unlimited food. For the first five years, the population skyrockets, doubling in size every few months. This is an example of:** A) Logistic growth B) Exponential growth C) Carrying capacity D) The bottleneck effect
 - 2. As a population of deer approaches the carrying capacity of its forest environment, which of the following is most likely to act as a density-dependent limiting factor to slow their growth?** A) A sudden forest fire B) An unusually harsh winter deep-freeze C) Starvation due to lack of grazing grass D) The amount of sunlight hitting the forest canopy
 - 3. In a standard terrestrial food web, which trophic level generally has the largest total physical biomass (the most organic matter by weight)?** A) Producers B) Primary Consumers C) Secondary Consumers D) Tertiary (Apex) Predators
 - 4. According to the 10% rule of energy transfer, what happens to the massive 90% of energy that is "lost" between trophic levels?** A) It is destroyed entirely. B) It is cycled back directly into the soil for plants to reuse. C) It is used for organismal metabolism and released largely as unrecoverable heat. D) It is stored permanently in the animal's fat.
 - 5. True or False:** The Carbon Cycle relies heavily on photosynthesis to pull carbon dioxide out of the atmosphere and package it into organic sugars. - ☐ True - ☐ False
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Reflection

- 6. Explain why top predators (like tigers, eagles, or killer whales) are always relatively rare in an ecosystem compared to the number of plants or herbivores.**

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7. Why must "Matter cycle" but "Energy flow" in an ecosystem? Briefly contrast how carbon atoms and sunlight behave differently as they enter the biosphere.

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8. Draw or describe the difference between the "J-curve" and the "S-curve" in population ecology. What specifically causes a J-curve to flatten into an S-curve?

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